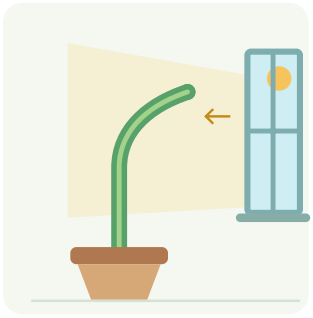
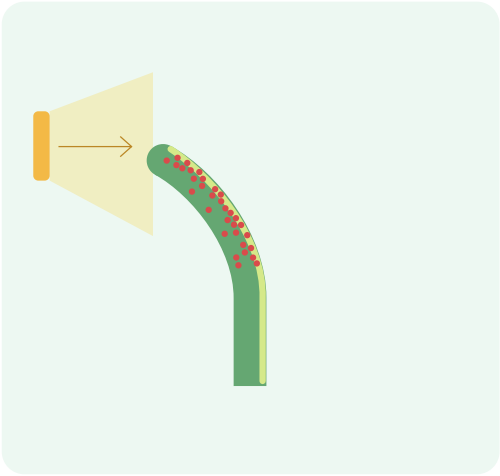
# Worksheet 3: Phototropic growth in plants

Name: _____ Class: _____ No.: _____

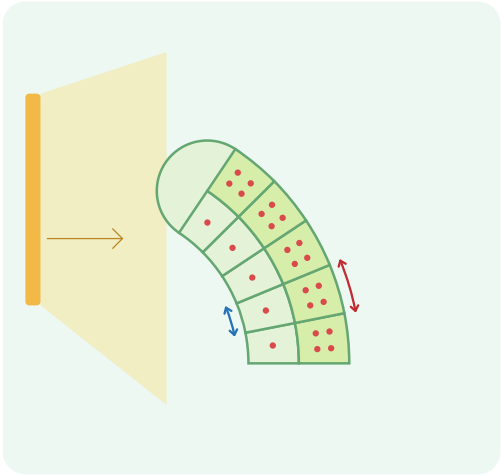
## Topic notes



- **Phototropism**（向光性） is growth in response to light direction. Shoots usually grow towards light (**positive phototropism**（向光性）). The left diagram shows a bending coleoptile（胚芽鞘）: part of the seedling's shoot that protects young leaves.
- The tip senses light from one side and contains **auxin**（生長素）, a chemical substance that is a **plant hormone**. More **auxin**（生長素） **passes down** the **shaded** side, stimulating cells in the elongation（延長） region to **elongate**（延長） and promote growth.
- Agar（瓊脂） allows chemical substances to pass through it.



Coleoptile（胚芽鞘）: more **auxin**（生長素） on the **shaded** right



Cell detail: more elongation（延長） on the right

Light from the left

↓

The right side is **shaded**

↓

More **auxin**（生長素） **passes down** the **shaded** side

↓

Cells on the right **elongate**（延長） more

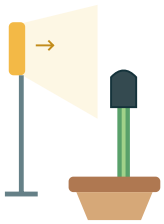
↓

The shoot bends left

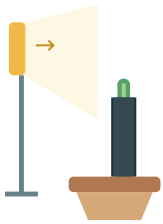
## Part 1 | Context and fair comparisons

Aim: investigate how shading position affects coleoptile（胚芽鞘） growth direction.

Design A · Light blocked at the tip



Design B · Light blocked below the tip



1. Identify the variables: tick the appropriate factors in each row. You may choose more than one.

| Variable type | Factors to choose from |
|---|---|
| Independent variable (the factor changed on purpose) | □ Time exposed to light □ The coleoptile（胚芽鞘）'s bending response □ Light direction and intensity □ Coleoptile（胚芽鞘） type, starting height and growth stage □ Light-blocking treatment □ Temperature and water supply |
| Dependent variable (the result measured or observed) | □ Time exposed to light □ The coleoptile（胚芽鞘）'s bending response □ Light direction and intensity □ Coleoptile（胚芽鞘） type, starting height and growth stage □ Light-blocking treatment □ Temperature and water supply |
| Controlled variables (the factors kept the same) | □ Time exposed to light □ The coleoptile（胚芽鞘）'s bending response □ Light direction and intensity □ Coleoptile（胚芽鞘） type, starting height and growth stage □ Light-blocking treatment □ Temperature and water supply |

### 2. What assumptions does this inquiry make?

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

Choose all suitable assumptions. You may choose more than one.

□ The coleoptiles（胚芽鞘） start in similar growth conditions.

□ The light-blocking cap and sleeve do not limit elongation（延長） or bending.

□ The treatments do not cause clear temperature differences between setups.

□ If the plants are the same type, their starting size does not need to be controlled.

### 3. Control group

To find out whether blocking light affects the coleoptile（胚芽鞘）'s growth direction, how would you design the control group?

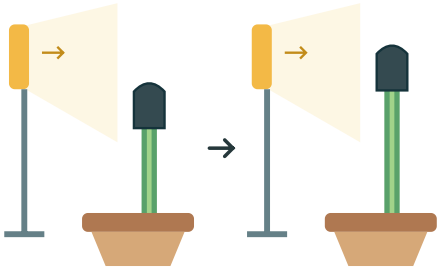
(Hint: The control group provides a baseline for comparison. It helps you confirm that a change is caused by the independent variable rather than other external factors, so the control group does not receive the independent variable treatment.)

## Part 2 | Results and analysis

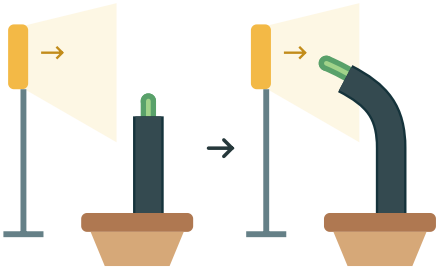
### 2.1 Main investigation: effect of shading on growth

Light is on the left. The sleeve covers below the exposed tip without restricting growth.

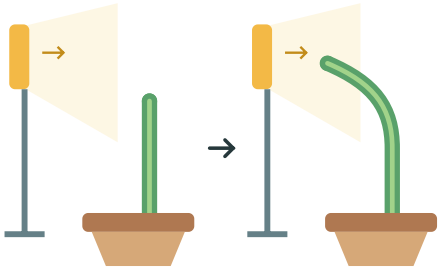
Setup A · Light blocked at the tip



Setup B · Light blocked below the tip



Setup C · No light blocked



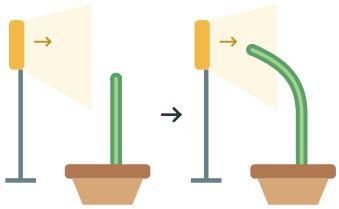
4. Record each setup's bending direction in the table.

| Setup | Bending direction |
|---|---|
| A | （Bends left ／ Bends right ／ No clear bending） |
| B | （Bends left ／ Bends right ／ No clear bending） |
| C | （Bends left ／ Bends right ／ No clear bending） |

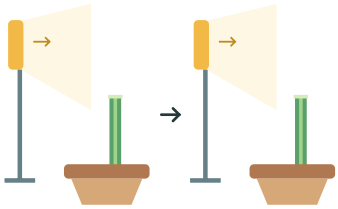
5. Compare C and A: after **blocking light at the tip**, what changes?
   （No clear bending towards the light ／ It shows the same bending towards light）
6. Compare C and B: when the lower part **does not** receive light directly, does the coleoptile（胚芽鞘） still bend?
   （It can still bend towards light ／ No clear bending towards the light）
7. Which part may **sense light from one side**?
   （Tip ／ Only the lower part）

### Extension investigation: effect of tip removal on growth

Setup C · No light blocked



Setup D · Tip removed



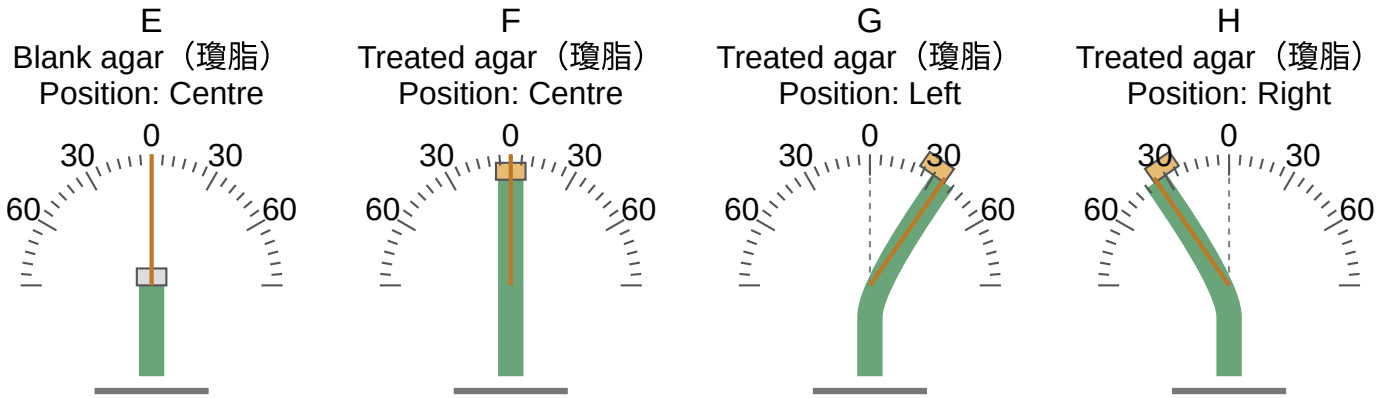
8. Record whether each setup **elongate**s（延長） in the table.

| Setup | Elongation（延長） |
|---|---|
| C | （Yes ／ No） |
| D | （Yes ／ No） |

9. What does comparing elongation（延長） in C and D support?
   □ The presence of the tip also affects the coleoptile（胚芽鞘）'s growth / elongation（延長）
   □ The presence of the tip does not affect the coleoptile（胚芽鞘）'s growth / elongation（延長）

## 2.3 Agar（瓊脂）

All cut-tip coleoptiles（胚芽鞘） start at similar sizes and stay in darkness. Blank agar（瓊脂） has not touched a tip; treated agar（瓊脂） has.




10. Record each setup's angle and bending direction.

| Setup | Angle (°) | Bending direction |
|---|---|---|
| E | _____ | （Bends left ／ Bends right ／ No clear bending） |
| F | _____ | （Bends left ／ Bends right ／ No clear bending） |
| G | _____ | （Bends left ／ Bends right ／ No clear bending） |
| H | _____ | （Bends left ／ Bends right ／ No clear bending） |

11. What does the comparison of E and F support?
（The tip may produce a growth-promoting effect that can be transferred ／ All agar（瓊脂） produces the same growth-promoting effect）

12. What does the comparison of F, G and H support?
（The position of treated agar（瓊脂） affects the direction of coleoptile（胚芽鞘） bending ／ The position of treated agar（瓊脂） does not affect the direction of coleoptile（胚芽鞘） bending）

13. When treated agar（瓊脂） is placed on the **left** side of a coleoptile（胚芽鞘） with its tip removed, compared with the right side, the left side receives （more ／ less） substance X, causing cells on the left to **elongate（延長）** （more ／ less） and the coleoptile（胚芽鞘） to bend to the （left ／ right） side.

## Part 3 | Integrated explanation

14. Use the experimental results and topic notes to explain why the coleoptile（胚芽鞘） in the diagram bends to the right.

